

A diet high in meat protein and potential renal acid load increases fractional calcium absorption and urinary calcium excretion without affecting m...

Our objective in this study was to determine the effects of a high-protein and high-potential renal acid load (PRAL) diet on calcium (Ca) absorption and retention and markers of bone metabolism. In a randomized crossover design, 16 postmenopausal women consumed 2 diets: 1 with low protein and low PRAL (LPLP; total protein: 61 g/d; PRAL: -48 mEq/d) and 1 with high protein and high PRAL (HPHP; total protein: 118 g/d; PRAL: 33 mEq/d) for 7 wk each separated by a 1-wk break. Ca absorption was measured by whole body scintillation counting of radio-labeled (47)Ca. Compared with the LPLP diet, the HPHP diet increased participants' serum IGF-I concentrations ($P < 0.0001$), decreased serum intact PTH concentrations ($P < 0.001$), and increased fractional (47)Ca absorption (mean $\pm$ pooled SD: 22.3 vs. $26.5 \pm 5.4\%$; $P < 0.05$) and urinary Ca excretion (156 vs. 203 ± 63 mg/d; $P = 0.005$). The net difference between the amount of Ca absorbed and excreted in urine did not differ between 2 diet periods (55 vs. 28 ± 51 mg/d). The dietary treatments did not affect other markers of bone metabolism. In summary, a diet high in protein and PRAL increases the fractional absorption of dietary Ca, which partially compensates for increased urinary Ca, in postmenopausal women. The increased IGF-I and decreased PTH concentrations in serum, with no change in biomarkers of bone resorption or formation, indicate a high-protein diet has no adverse effects on bone health.